

Labo 08.02 – AlwaysOn Availability Groups

Volledige Stappenplan

Databankbeheer Dany De Deken

Academiejaar: 2025-2026

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In deze handleiding worden de volgende onderdelen uitgevoerd

- *Een server toevoegen aan een domein*
- *Een SQL server op een Windows Server Core installeren en configureren*
- *De juiste firewall regels aanmaken op beide Member servers*
- *Configureren van een Windows Server Failover Cluster*
- *Always On Availability Groups inschakelen en configureren op beide member servers.*
- *Een Availability Group opzetten voor een database*
- *Synchronisatie & status-controle*

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Vorbereiding

Onderstaande zaken moeten zeker gebeurd zijn voordat je verder kunt gaan.

- Een Domaincontroller
- Basis configuratie zoals hostname en juist IP-adress op alle toestellen.
- Zorg dat beide servers in hetzelfde domein zitten. Als dit niet het geval is zie pagina 3.
- SQL Server Developer edition 2022 geïnstalleerd op beide Member servers.
- SQL Server installeren op Member02 (core). Als dit niet het geval is zie pagina 7.
- Controleer dat je domeinaccount sysadmin is op beide SQL-instances.
- De Member servers moeten toegevoegd zijn met een Managed-Service-Account op de Domaincontroller.
- Coreccte mappenstructuur op beide Member servers:

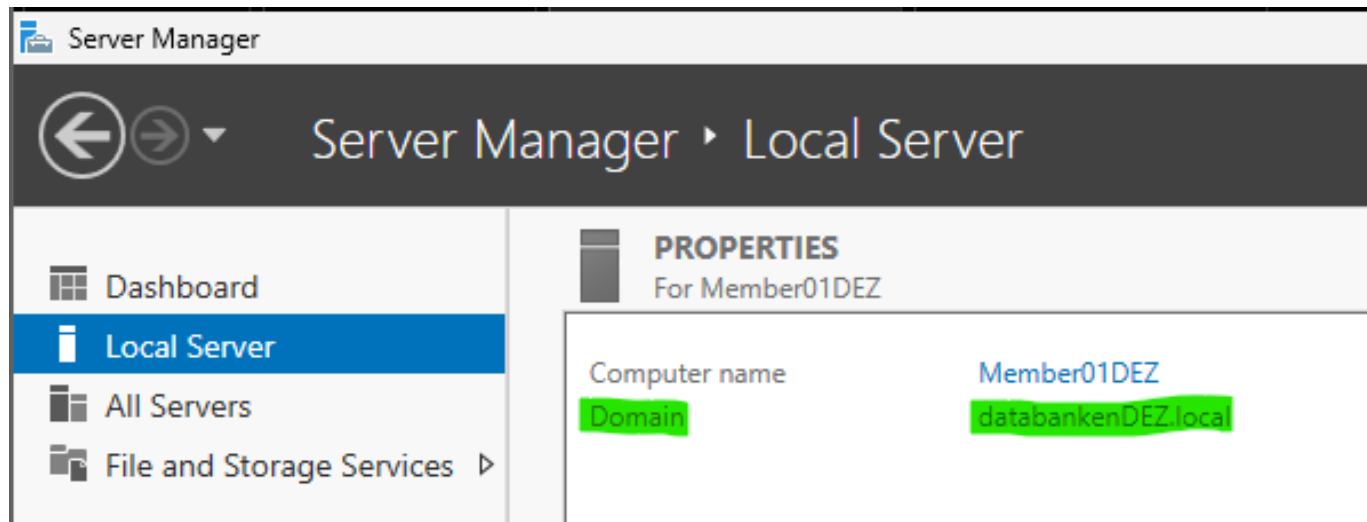
- \SQLBackup
- \SQLData
- \SQLLogs
- \tmp

Member servers toevoegen aan een domein

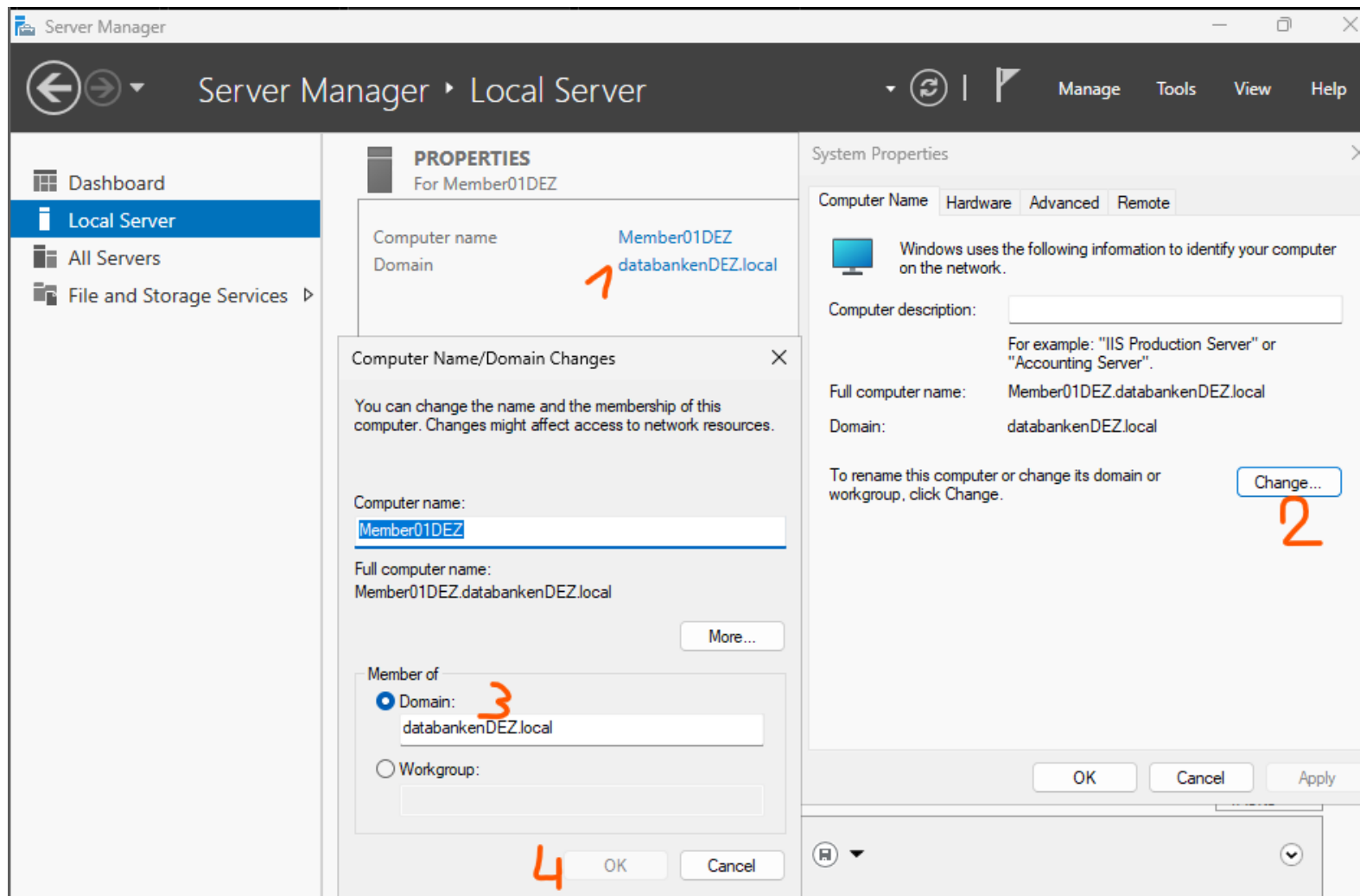
Op de GUI server:

ga naar server manager, local server

Controleer of die in het juiste domein zit.



Als dit niet het geval is, volg dan deze stappen om die aan een domein toe te voegen:



Op de Core server

Controleer of die in het juiste domein zit:

```
=====
Welcome to Windows Server 2025 Standard
=====

1) Domain/workgroup:          Domain: databankenDEZ.local
2) Computer name:            MEMBER02DEZ
3) Add local administrator
4) Remote management:        Enabled
```

Als dit niet het geval is, volg dan deze stappen om die in een domein te plaatsen:

- Ga naar PowerShell:

```
15) Exit to command line (PowerShell)

Enter number to select an option: 15
```

-Voer dit commando uit met de juiste naam van jouw domein:

Add-Computer -DomainName "databankendez.local" -Credential databankendez\administrator

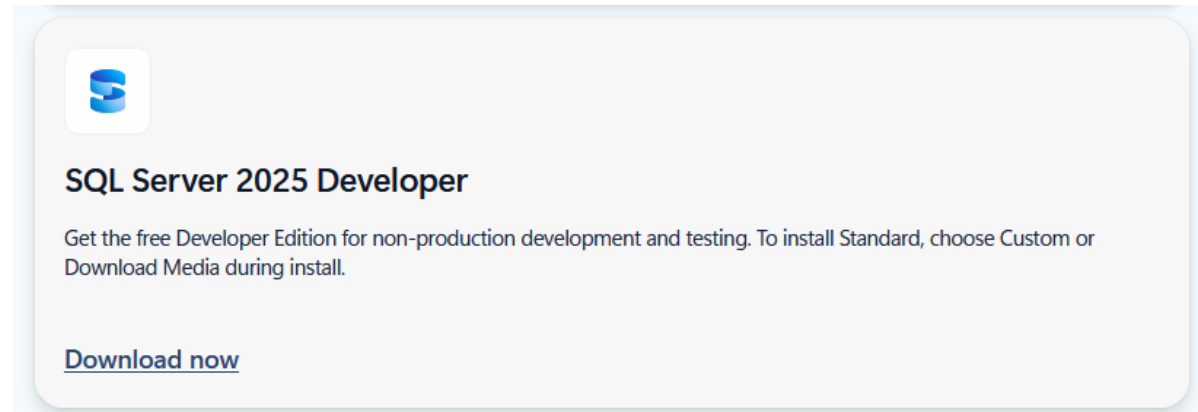
-Je moet dan inloggen met de wachtwoord.

-Start de server opnieuw op om de wijzigingen uit te voeren: Restart-Computer

Installatie SQL Server op de Core server

Download op je eigen laptop de SQL Server Developer Edition via deze link:

<https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

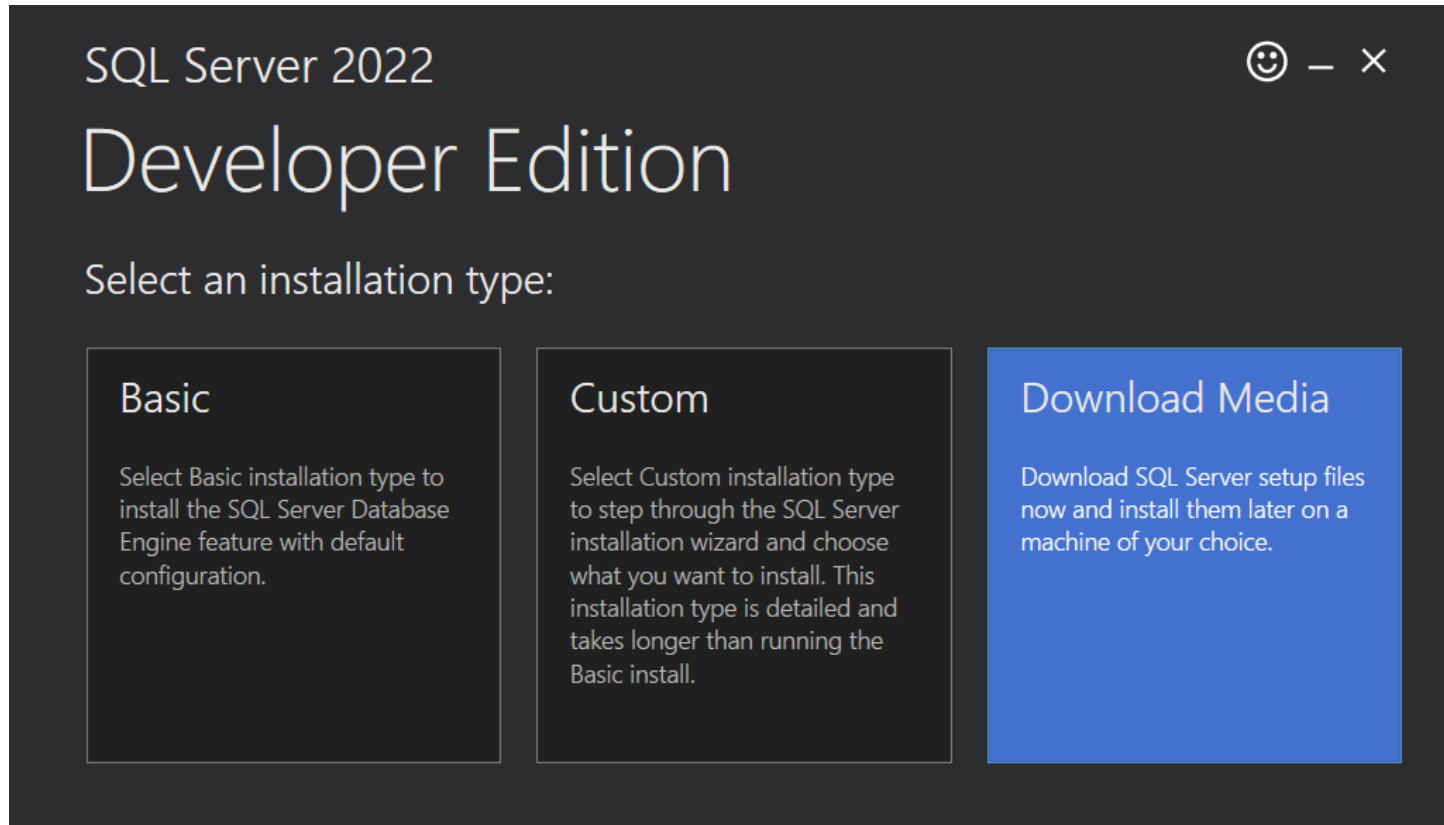


Zoals je ziet staat er alleen de 2025 versie. Om de 2022 versie te installeren, bezoek deze link van Microsoft:

<https://download.microsoft.com/download/c/c/9/cc9c6797-383c-4b24-8920-dc057c1de9d3/SQL2022-SSEI-Dev.exe>

Als het klaar is met downloaden krijg je de installer te zien.

Selecteer Download Media.



Kies een taal en selecteer ISO. Kies dan een plek waar je de ISO wilt opslaan.

Specify SQL Server installer download

SELECT LANGUAGE

English

WHICH PACKAGE WOULD YOU LIKE TO DOWNLOAD?

- ☒ ISO (1207 MB)
Uncompressed, mountable disk image media
- ☐ CAB (1290 MB)
Compressed media, .exe and .box files

[Or, go to the Microsoft Azure portal to provision SQL Server in the cloud](#)

SELECT DOWNLOAD LOCATION *:

C:\Users\r1ffi\Downloads

 Browse

Klik dan op download. Dit kan even duren.

Nu heb je de ISO. Dit is het bestand dat je op de Server Core moet zetten via net use.

Controleer je lokale gebruikersnaam op je laptop
Open CMD en typ: whoami

Je krijgt zoiets als:
msi\dina

Gebruik exact die info op de server core om de ISO erop te krijgen, zoals deze voorbeeld:
net use Z: \\msi\sql2025 /user:msi\dina <jouw-windows-wachtwoord>

Als het goed is krijg je:
The command completed succesfully

Nu is het de bedoeling dat je de ISO mount op de Core Server met deze commando:
powershell mount-diskimage -imagepath "Z:\SQLServer2025-x64-ENU-EntDev.iso"

SQL Server installeren

Ga naar de drive, in dit geval Z:

Z:

cd setup

Maak directory's voor SQL data, logs en backups:

mkdir "K:\SQLData"

mkdir "K:\SQLLogs"

mkdir "K:\SQLBackups"

Voer de installatie uit:

```
Z:\setup.exe /QS /ACTION=Install /FEATURES=SQLENGINE /INSTANCENAME=MSSQLSERVER /SQLSVCACCOUNT="NT  
AUTHORITY\NETWORK SERVICE" /SQLSYSADMINACCOUNTS="Administrator" /SQLUSERDBDIR="K:\SQLData"  
/SQLUSERDBLOGDIR="K:\SQLLogs" /SQLBACKUPDIR="K:\SQLBackups" /IAcceptSQLServerLicenseTerms
```

Wacht tot de installatie klaar is, je kunt dan controleren met:

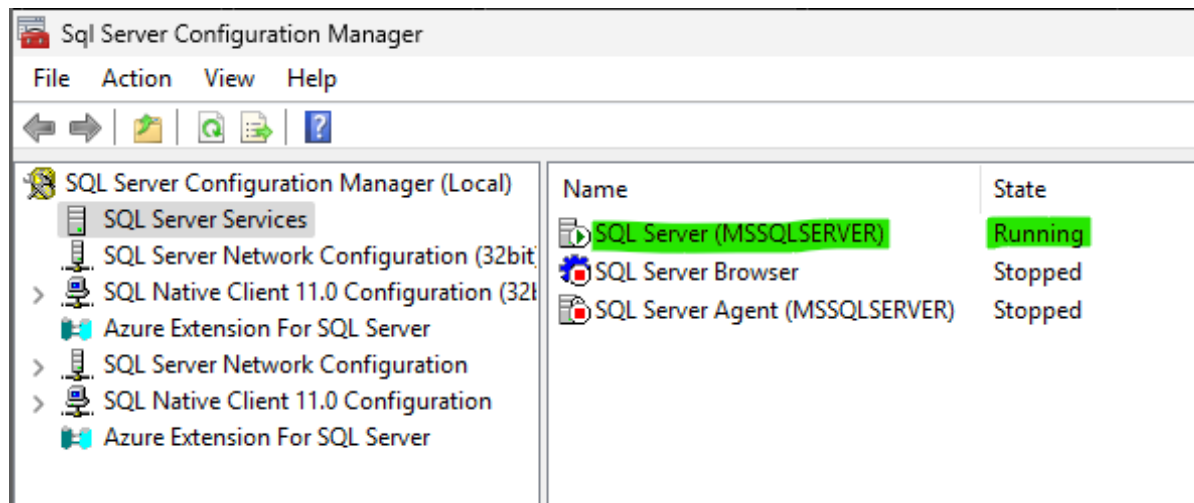
Get-Service *sql*

Moet RUNNING zijn.

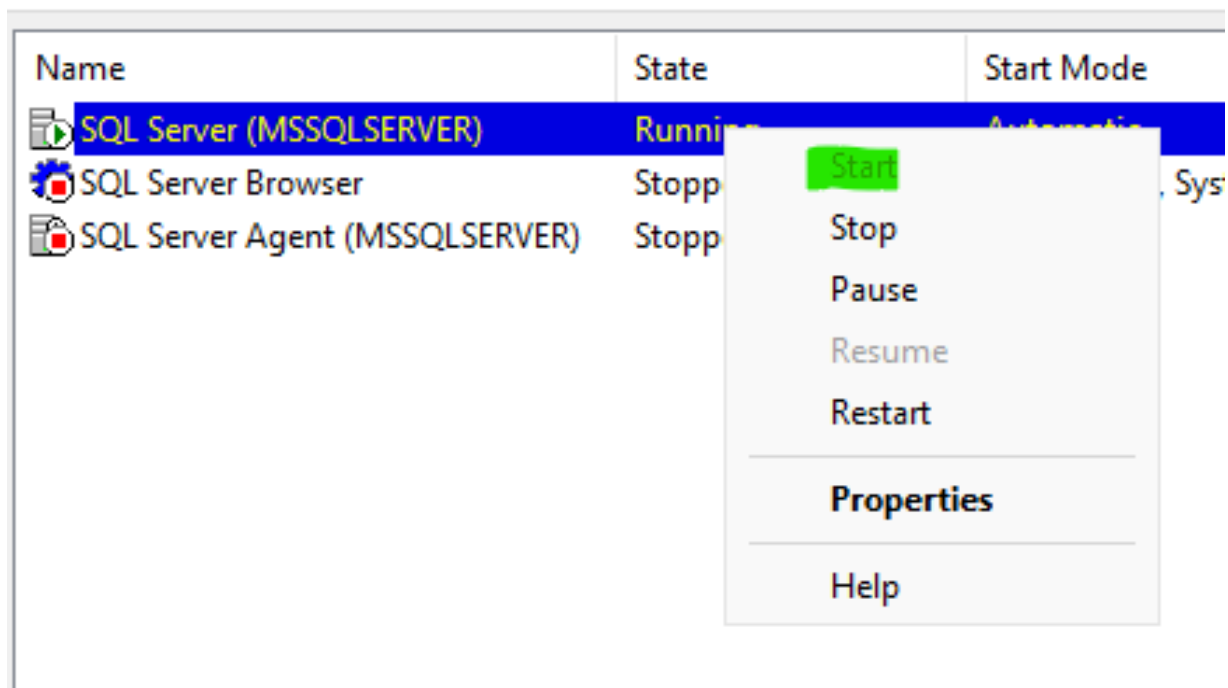
Nu heb je SQL succesvol geïnstalleerd.

Controle SQL configuratie op Member01DEZ (GUI)

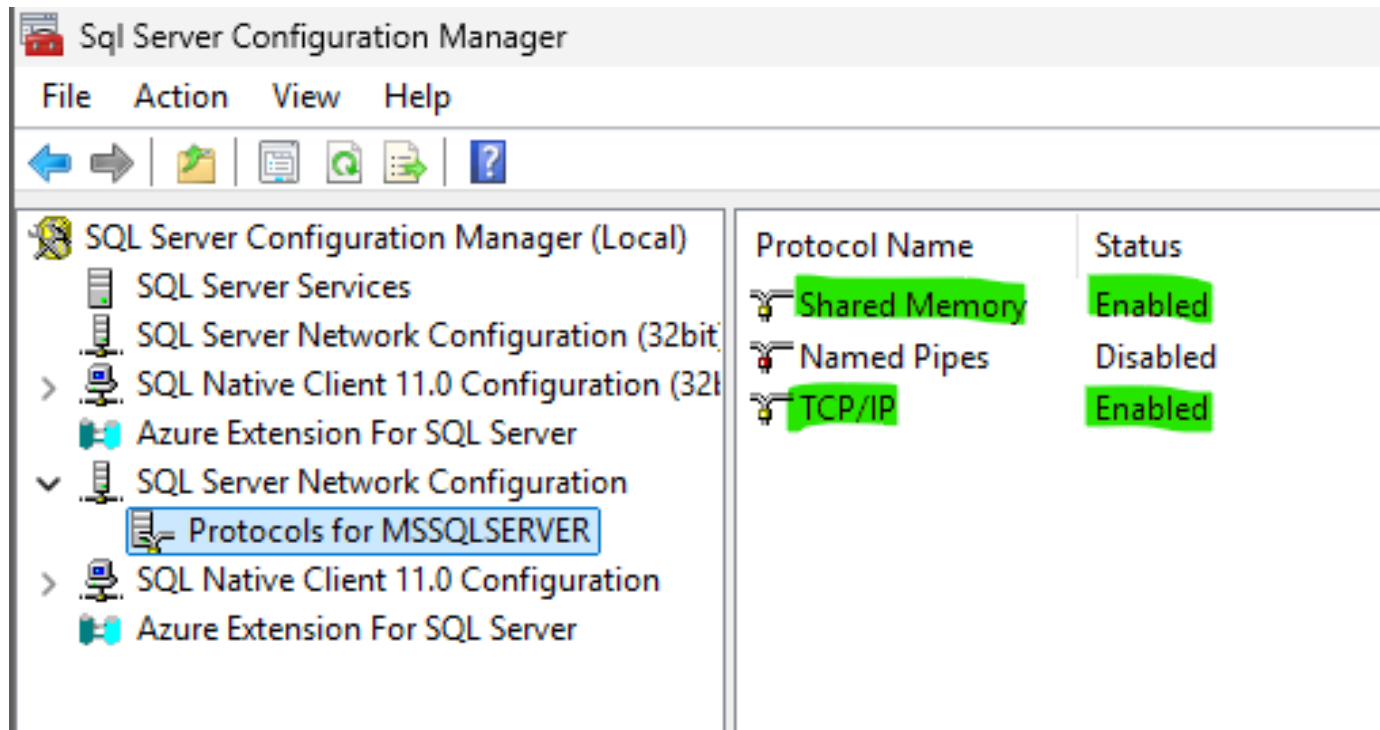
- Open SQL Server Configuration Manager.
- Controleer of SQL Server service draait. "Running"



Als deze niet RUNNING is. Klik dan met je rechtermuisknop op SQL Server en klik op start:



Controleer TCP/IP Enabled onder Protocols en ook Shared Memory:



Als dit niet het geval is → rechtermuisknop → Enable

SQL configuratie op Member02DEZ (Core)

Controleer dat TCP/IP Enabled onder Protocols

Powershell Commando:

Get-NetTCPConnection -LocalPort 1433

```
PS E:\> Get-NetTCPConnection -LocalPort 1433
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
-----	-----	-----	-----	-----	-----	-----
::	1433	::	0	Listen		3256
0.0.0.0	1433	0.0.0.0	0	Listen		3256

Start SQL daarna opnieuw op:

Restart-Service MSSQLSERVER

Firewall configuratie

Op de Core server:

Gebruik onderstaande commando's in Powershell.

SQL-poort openen:

```
New-NetFirewallRule -DisplayName "SQL 1433" -Direction Inbound -Protocol TCP -LocalPort 1433 -Action Allow
```

AlwaysOn endpoint-poort openen (5022):

```
New-NetFirewallRule -DisplayName "SQL AG Endpoint 5022" -Direction Inbound -Protocol TCP -LocalPort 5022 -Action Allow
```

Firewall configuratie op GUI server:

Open Powershell met Administrator rechten en voer onderstaande commando's in:

```
New-NetFirewallRule -DisplayName "SQL AG Endpoint 5022" -Direction Inbound -Protocol TCP -LocalPort 5022 -Action Allow
```


Windows-features installeren op beide servers voor Failover Cluster

Op de GUI Server

open Powershell als Administrator en voer onderstaand commando's uit:

Install-WindowsFeature -Name Failover-Clustering -IncludeManagementTools

Get-WindowsFeature Failover-Clustering

Restart-Computer -Force

```
PS C:\Users\Administrator> Install-WindowsFeature -Name Failover-Clustering -IncludeManagementTools
```

Success	Restart Needed	Exit Code	Feature Result
True	No	Success	{Failover Clustering, Remote Server Admini...

```
PS C:\Users\Administrator> Get-WindowsFeature Failover-Clustering
```

Display Name	Name	Install State
[X] Failover Clustering	Failover-Clustering	Installed

```
PS C:\Users\Administrator> |
```

Op de Core Server

Voer onderstaande commando's uit in PowerShell:

Install-WindowsFeature -Name Failover-Clustering

Get-WindowsFeature Failover-Clustering

Restart-Computer -Force

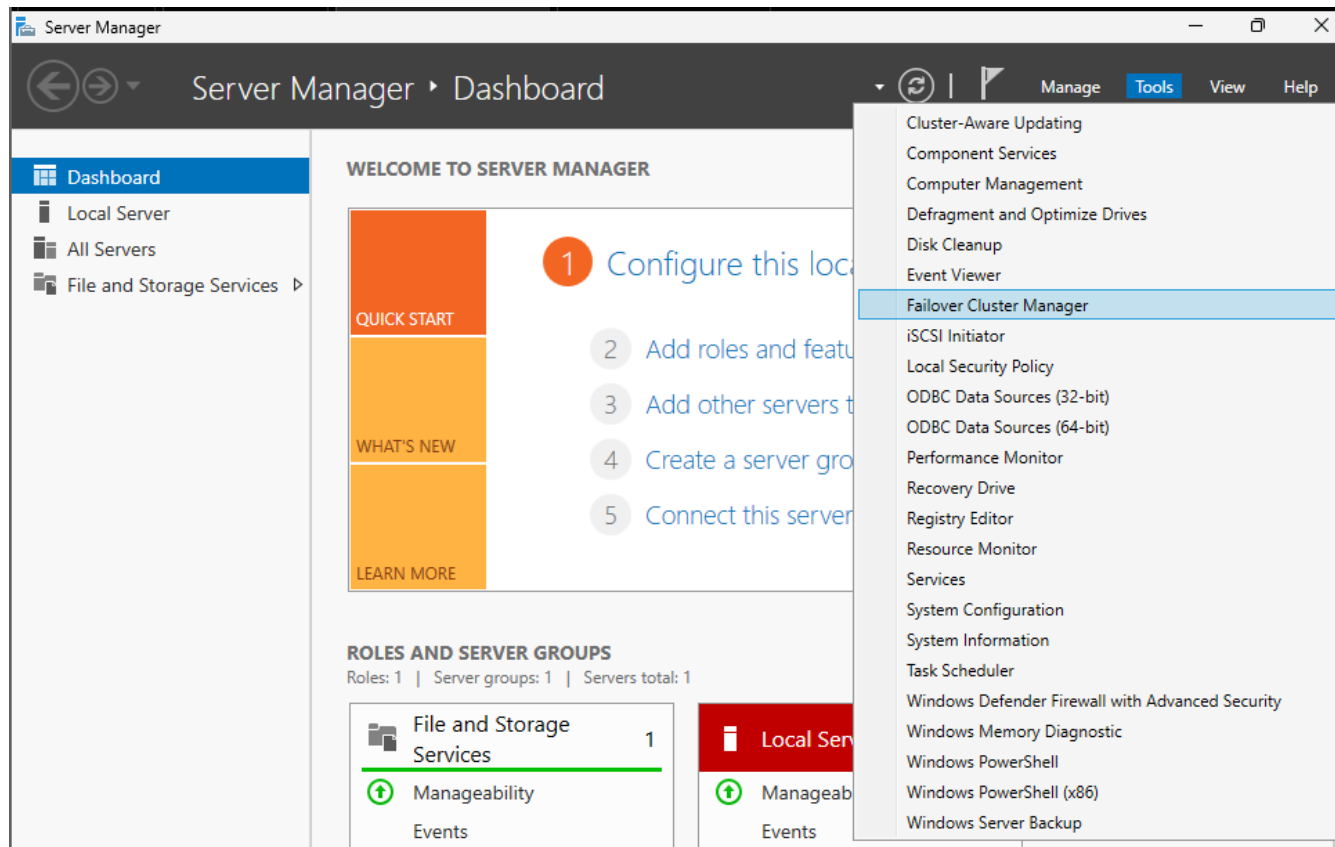
```
PS C:\Users\Administrator> Get-WindowsFeature Failover-Clustering
```

Display Name	Name	Install State
-----	----	-----
[] Failover Clustering	Failover-Clustering	Available

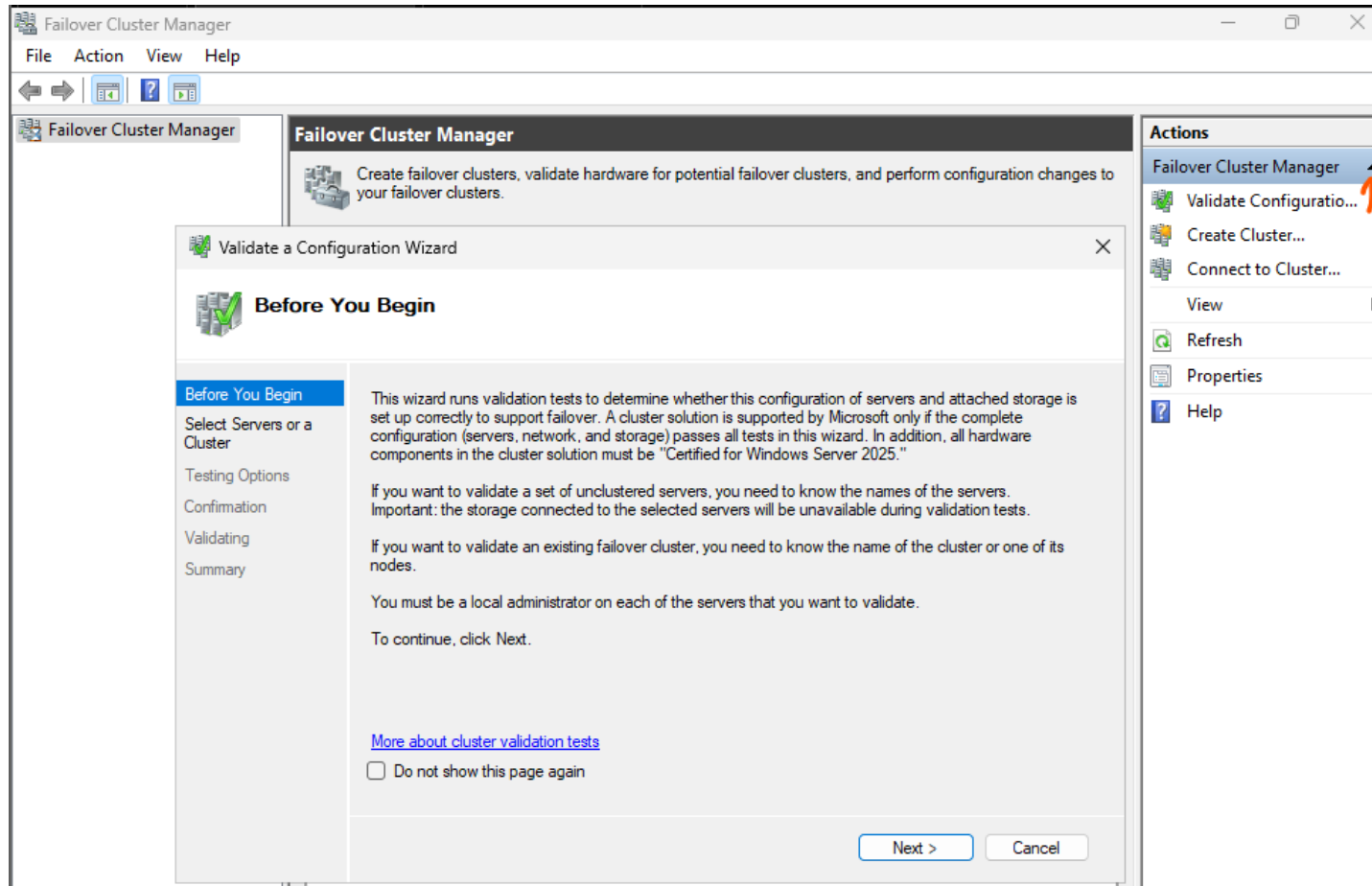
Failover Cluster maken

Op de GUI server

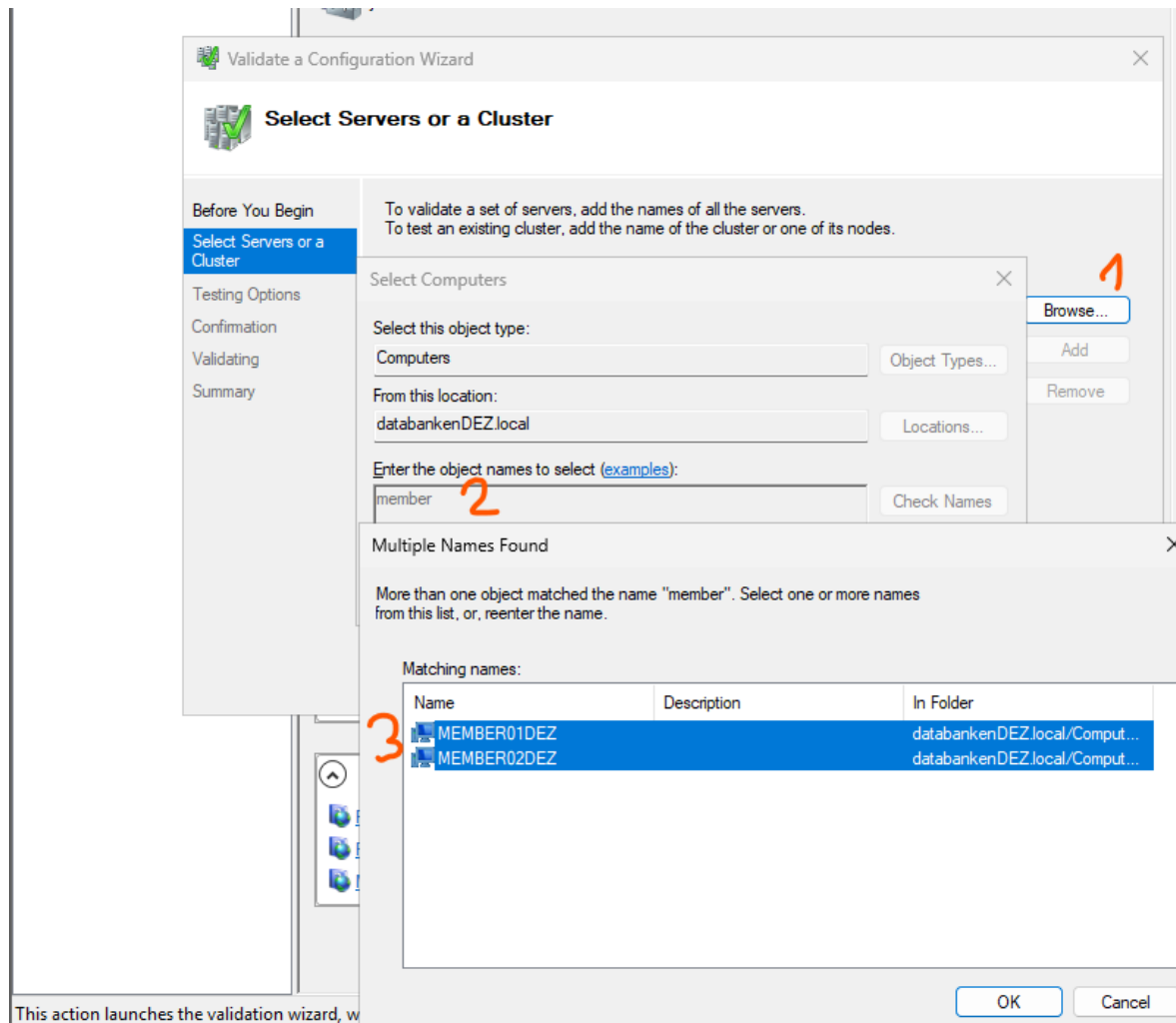
Ga naar Server Manager → Tools → Failover Cluster Manager



Klik op Validate Configuration om in de wizard te komen. Klik dan Next.



Nu moet je de Member servers selecteren door eerst op Browse te klikken. Dan typ je “member” en dan enter. Selecteer nu beide servers. En klik op OK.



Klik dan op Next.

Bij de Testing options kies je voor run all tests:

Validate a Configuration Wizard

Testing Options

Before You Begin
Select Servers or a Cluster
Testing Options
Confirmation
Validating
Summary

Choose between running all tests or running selected tests.

The tests examine the Cluster Configuration, Hyper-V Configuration, Inventory, Network, Storage, and System Configuration.

Microsoft supports a cluster solution only if the complete configuration (servers, network, and storage) can pass all tests in this wizard. In addition, all hardware components in the cluster solution must be "Certified for Windows Server 2025."

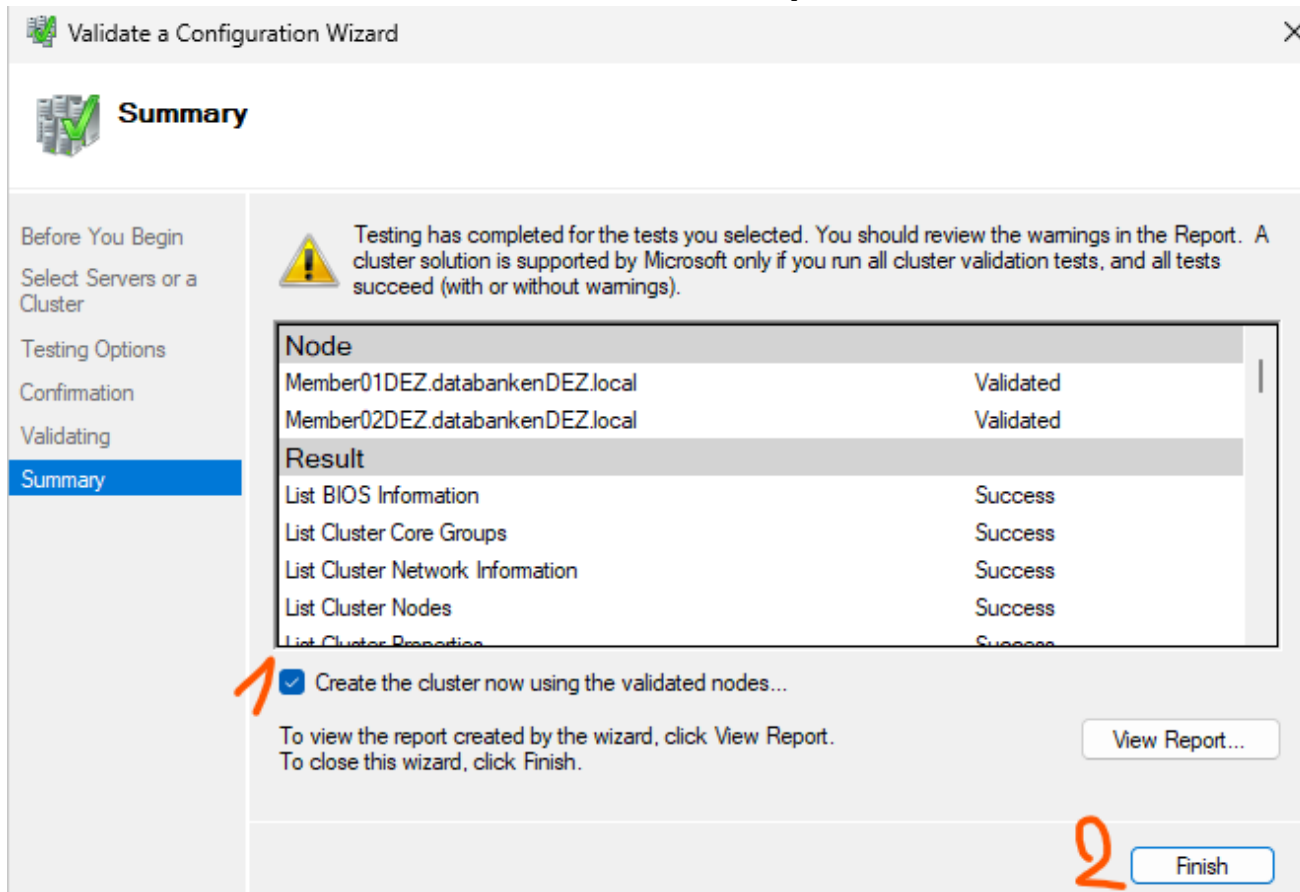
1 ☒ Run all tests (recommended)
☐ Run only tests I select

[More about cluster validation tests](#)

2 < Previous Next > Cancel

Klik dan 2x op Next om de tests te starten. Het testen kan even duren.

Als het testen klaar is selecteer dan onderstaande vak en klik op finish:



Validate a Configuration Wizard

Summary

Testing has completed for the tests you selected. You should review the warnings in the Report. A cluster solution is supported by Microsoft only if you run all cluster validation tests, and all tests succeed (with or without warnings).

Node	
Member01DEZ.databankenDEZ.local	Validated
Member02DEZ.databankenDEZ.local	Validated

Result	
List BIOS Information	Success
List Cluster Core Groups	Success
List Cluster Network Information	Success
List Cluster Nodes	Success
List Cluster Properties	Success

☒ Create the cluster now using the validated nodes...

To view the report created by the wizard, click View Report.
To close this wizard, click Finish.

[View Report...](#)

[Finish](#)

Dan kom je automatisch in de Create Cluster Wizard terecht. Geef de cluster een naam en een ip address in de netwerk range die je hebt. De ip address mag nog niet in gebruik zijn.

Create Cluster Wizard

Access Point for Administering the Cluster

Before You Begin

Access Point for Administering the Cluster

Confirmation

Creating New Cluster

Summary

Type the name you want to use when administering the cluster.

Cluster Name: 1 SQLcluster

The NetBIOS name is limited to 15 characters. One or more IPv4 addresses could not be configured automatically. For each network to be used, make sure the network is selected, and then type an address.

	Networks	Address
☒	172.25.87.0/24	172.25.87.200 2

3

< Previous Next > Cancel

Check of het vakje aangevinkt staat en klik dan op next:

Create Cluster Wizard

Confirmation

Before You Begin
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary

You are ready to create a cluster.
The wizard will create your cluster with the following settings:

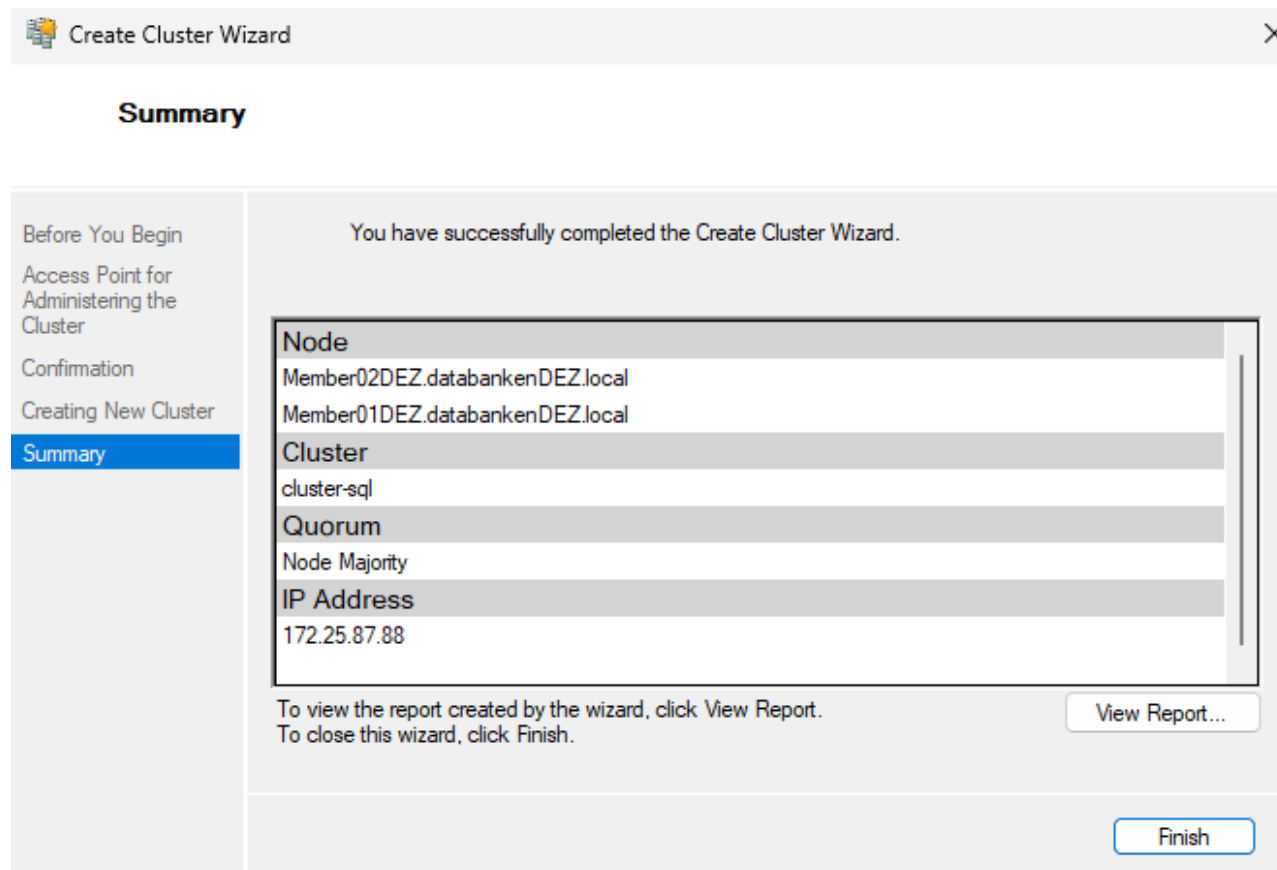
Cluster
SQLcluster
Node
Member02DEZ.databankenDEZ.local
Member01DEZ.databankenDEZ.local
Cluster registration
DNS and Active Directory Domain Services

☒ Add all eligible storage to the cluster.
To continue, click Next.

< Previous **Next >** Cancel

De cluster wordt nu aangemaakt. Dit kan even duren.

Als de cluster succesvol is aangemaakt klik op Finish:



Cluster Quorum Domain Controller

Het is nu de bedoeling dat de Domain Controller de baas wordt.

Hiervoor maken we een gedeelde map aan op DC. Dit heet de File Share witness.

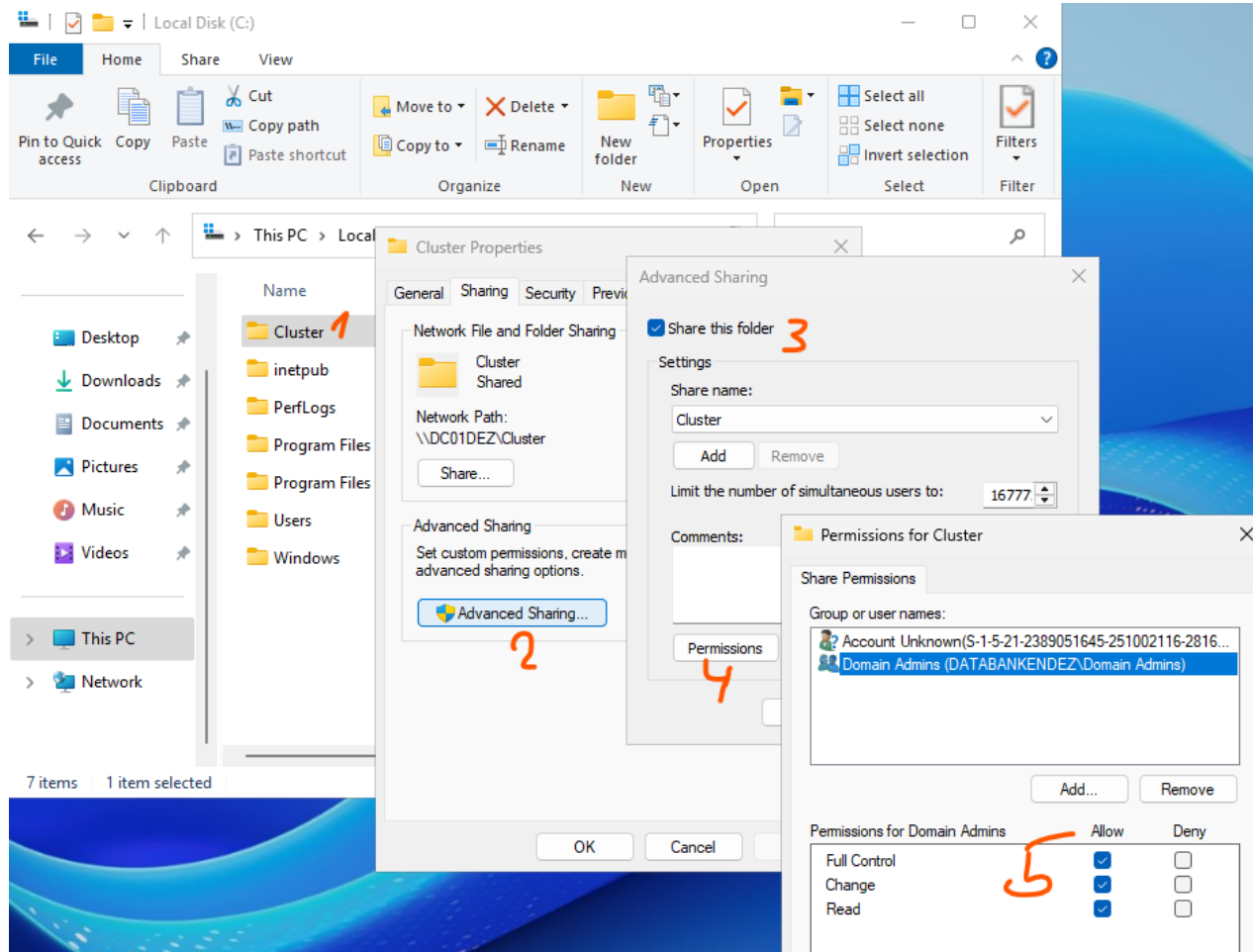
Maak een folder "Cluster" op de DC en deel die met Full-Access rechten aan de Admins:

`New-SmbShare -Name "ClusterWitness" -Path "C:\Cluster" -FullAccess "cluster-sql$", "Domain Admins"`

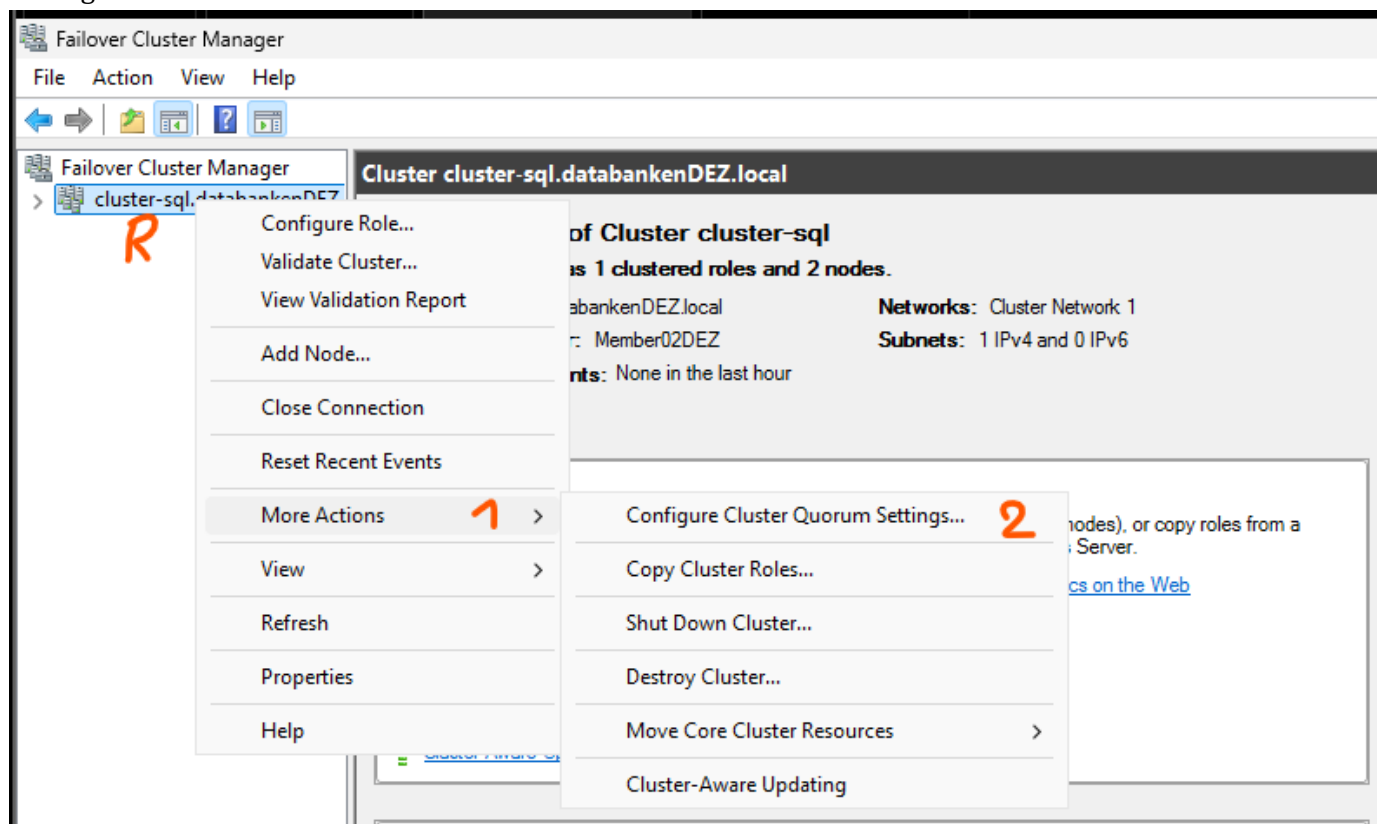
```
PS C:\Users\Administrator> New-SmbShare -Name "ClusterWitness" -Path "C:\Cluster" -FullAccess "cluster-sql$", "Domain Admins"
```

Name	ScopeName	Path	Description
----	-----	----	-----
ClusterWitness	*	C:\Cluster	

Je kunt dit ook handmatig doen:



Ga nu terug naar Member01 → Failover Cluster Manager → Rechtermuisknop op de cluster → More Actions → Configure cluster quorum settings



Selecteer de middelste optie en klik op Next:

Configure Cluster Quorum Wizard

Select Quorum Configuration Option

Before You Begin

Select Quorum Configuration Option

Select Quorum Witness

Confirmation

Configure Cluster Quorum Settings

Summary

Select a quorum configuration for your cluster.

☐ Use default quorum configuration
The cluster determines quorum management options, including the quorum witness.

☒ Select the quorum witness
You can add or change the quorum witness. The cluster determines the other quorum management options.

☐ Advanced quorum configuration
You determine the quorum management options, including the quorum witness.

[Failover Cluster Quorum and Witness Configuration Options](#)

< Previous Next > Cancel

Selecteer dan de 2de optie om een file share witness te configureren en klik op Next:

Configure Cluster Quorum Wizard

Select Quorum Witness

Before You Begin
Select Quorum Configuration Option
Select Quorum Witness
Configure File Share Witness
Confirmation
Configure Cluster Quorum Settings
Summary

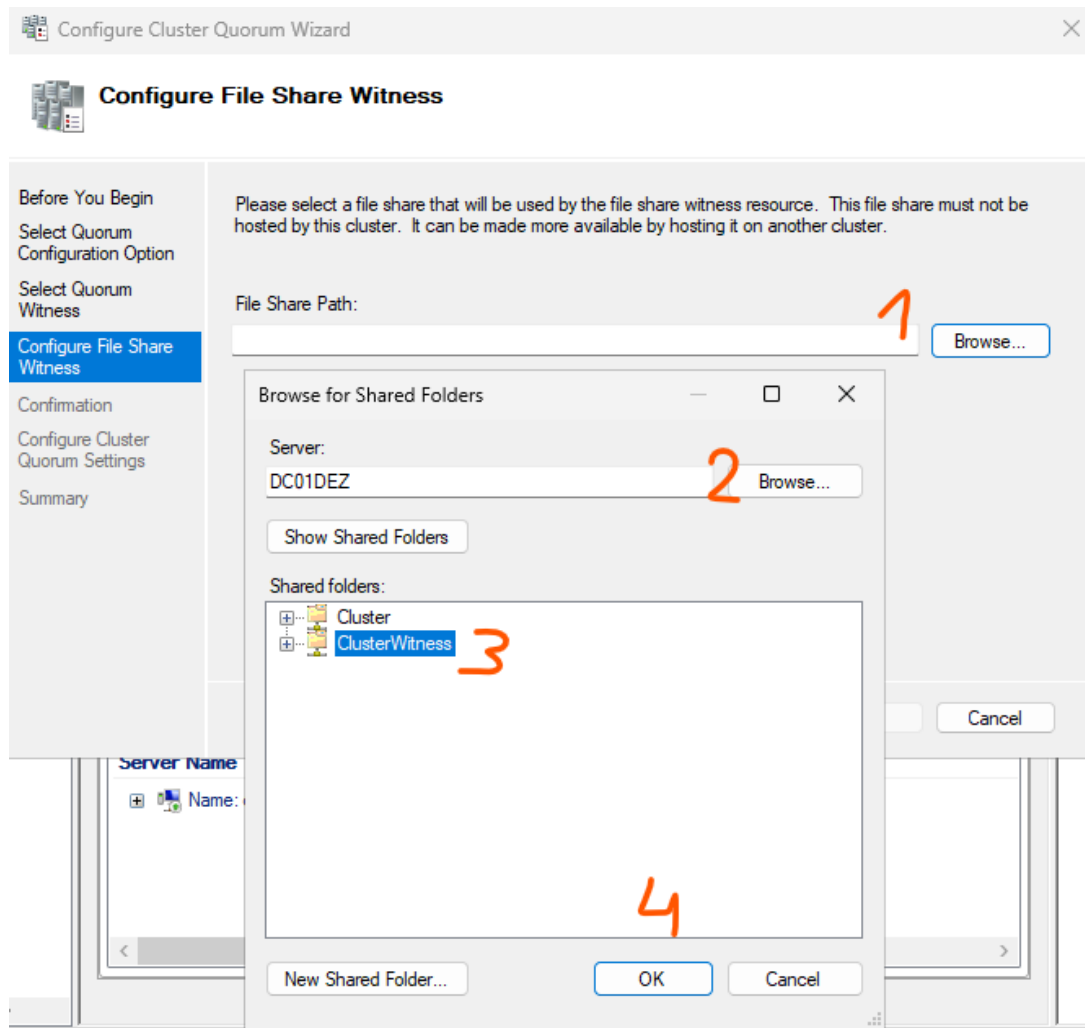
Select a quorum witness option to add or change the quorum witness for your cluster configuration. As a best practice, configure a quorum witness to help achieve the highest availability of the cluster.

- ☐ Configure a disk witness
Adds a quorum vote of the disk witness
- ☒ **Configure a file share witness**
Adds a quorum vote of the file share witness
- ☐ Configure a cloud witness
Adds a quorum vote of the cloud witness
- ☐ Do not configure a quorum witness

[Failover Cluster Quorum and Witness Configuration Options](#)

< Previous **Next >** Cancel

Voeg de DC toe als server en selecteer de juiste map die je hiervoor hebt gemaakt:



Klik dan op next.

Op het einde klik je op Finish:

 Configure Cluster Quorum Wizard ✕

Summary

Before You Begin

Select Quorum Configuration Option

Select Quorum Witness

Configure File Share Witness

Confirmation

Configure Cluster Quorum Settings

Summary

You have successfully configured the quorum settings for the cluster.

Cluster Managed Voting
Enabled
Witness Type
File Share Witness
Witness Resource
\\DC01DEZ\\ClusterWitness

To view the report created by the wizard, click View Report.
To close this wizard, click Finish.

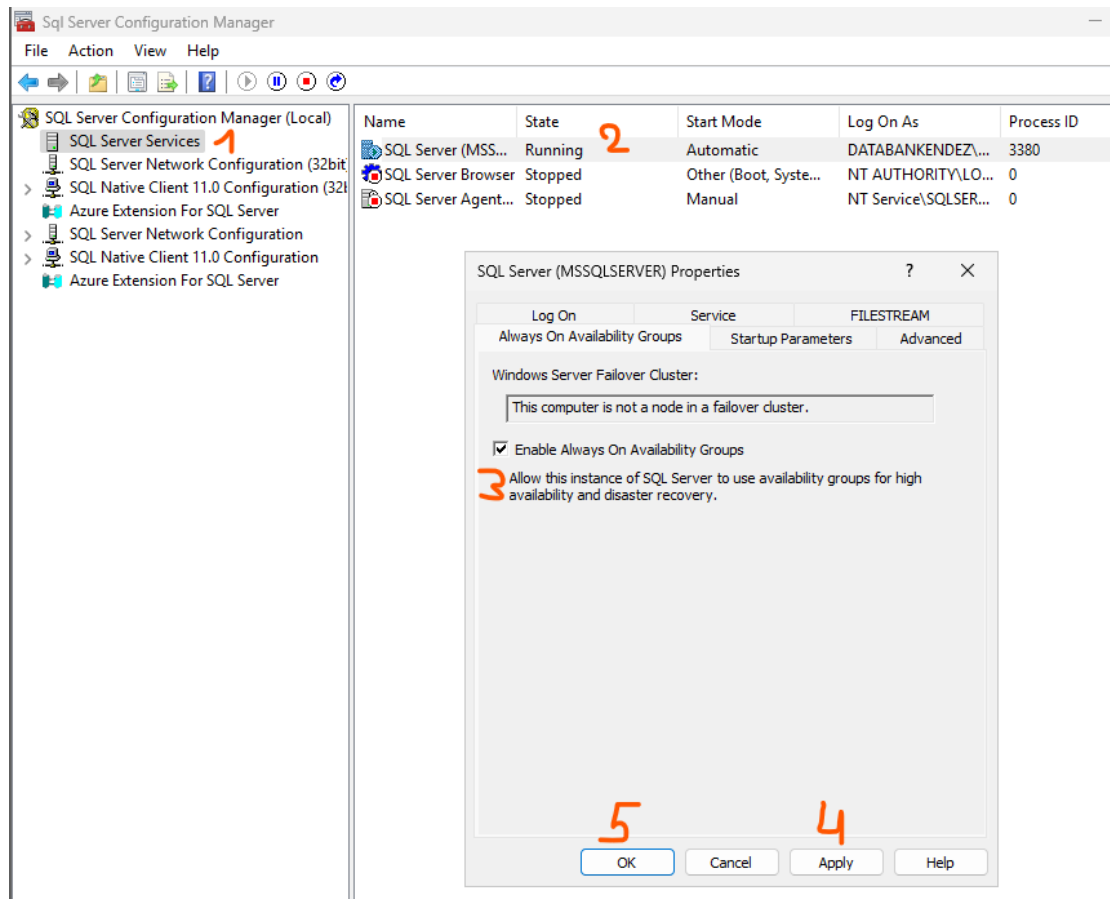
View Report...

Finish

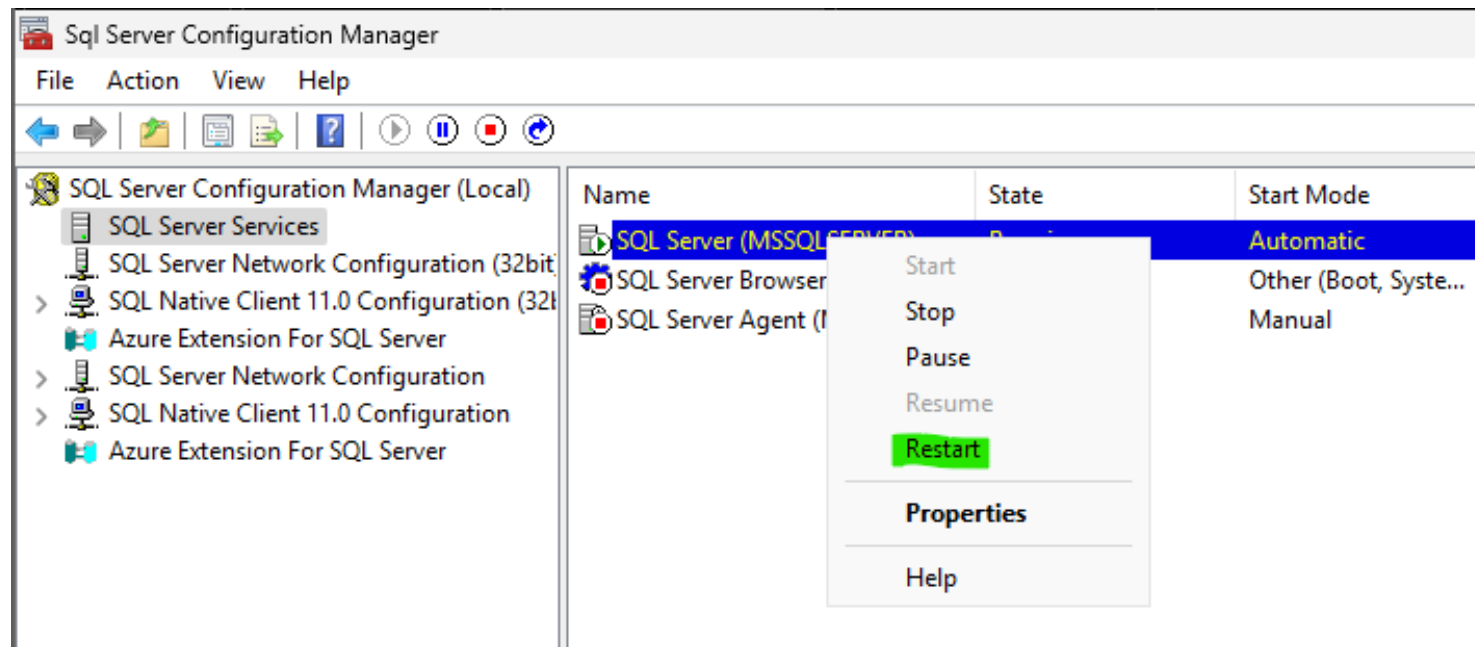
AlwaysOn inschakelen

GUI

SQL Server Configuration Manager → SQL Server Services → Properties → AlwaysOn High Availability → Enable.



Als je de wijziging hebt opgeslagen moet je SQL herstarten! Rechtermuisknop → Restart:



Op de Core server

Gebruik deze commando's in Powershell:

```
Import-Module SQLPS
```

```
Enable-SqlAlwaysOn -ServerInstance "MEMBER02DEZ" -Force
```

```
Restart-Service MSSQLSERVER
```

Controleer met deze commando of het is enabled, de waarde moet 1 zijn:

```
Invoke-Sqlcmd -ServerInstance "MEMBER02DEZ" -Query "SELECT SERVERPROPERTY('IsHadrEnabled') AS IsHadrEnabled;"
```

```
PS C:\Users\Administrator> Invoke-Sqlcmd -ServerInstance "MEMBER02DEZ" -Query "SELECT SERVERPROPERTY('IsHadrEnabled') AS IsHadrEnabled;"
IsHadrEnabled
-----
1
```

Open SQL Server van Member01. Op je Member01 zelf of op de client.

Connect to Server

SQL Server

Login | Connection Properties | Always Encrypted | Additional Connection Parameters

Server

Server type: Database Engine

Server name: MEMBER01DEZ

Authentication: Windows Authentication

User name: MEMBER01DEZ\Administrator

Password:

☐ Remember password

Connection Security

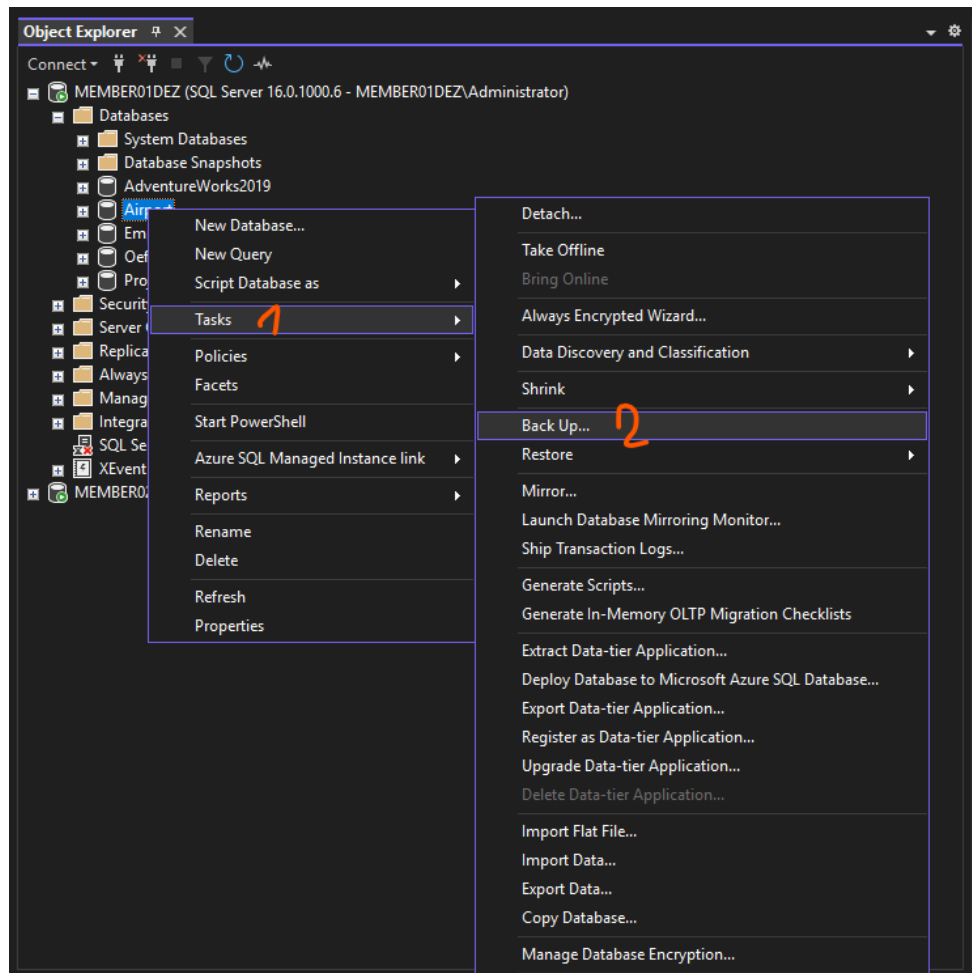
Encryption: Mandatory

☒ Trust server certificate

Host name in certificate:

Connect Cancel Help Options <<

Klik op de database “Airport” → Rechtermuisknop → Tasks → Back Up...



Zorg ervoor dat de Backup type “full” is:

Back Up Database - Airport

Select a page: General, Media Options, Backup Options

Script Help

Source

Database: Airport

Recovery model: FULL

Backup type: Full

☐ Copy-only backup

Backup component:

☒ Database

☐ Files and filegroups:

Destination

Back up to: Disk

C:\backup\airport_resync.bak

Add... Remove Contents

Connection

Server: Member01DEZ

Connection: MEMBER01DEZ\Administrator

[View connection properties](#)

Progress

Ready

OK Cancel

Open daarna een nieuwe query in Member01 en voer onderstaande query uit voor de endpoint:

```
USE master;
```

```
GO
```

```
Changed database context to 'master'.
```

```
IF NOT EXISTS (SELECT 1 FROM sys.endpoints WHERE name = 'Hadr_endpoint')
```

```
EXEC ('CREATE ENDPOINT Hadr_endpoint
```

```
STATE - STARTED
```

```
AS TCP (LISTENER_PORT = 5022)
```

```
FOR DATA MIRRORING (
```

```
ROLE - ALL,
```

```
AUTHENTICATION = WINDOWS NEGOTIATE,
```

```
ENCRYPTION = REQUIRED ALGORITHM AES
```

```
]);
```

```
IF NOT EXISTS (SELECT 1 FROM sys.server_principals WHERE name = 'databankenDEZ\sql')
```



```
CREATE LOGIN [databankenDEZ\sq1] FROM WINDOWS;
```

```
GRANT CONNECT ON ENDPOINT :: Hadr_endpoint TO [databankenDEZ\sq1];
```

```
GO
```

```
SELECT name, state_desc, port
```

```
FROM sys.tcp_endpoints
```

```
WHERE type_desc = 'DATABASE_MIRRORING';
```

```
go
```

The screenshot shows the SQL Server Enterprise Manager interface. The main window displays a SQL script in a dark-themed editor. The script is for creating a data mirroring endpoint and a server principal. The script includes comments and uses standard SQL syntax. The bottom status bar indicates 'No issues found' and shows the current line and column numbers. The bottom pane shows the 'Results' tab with a table of endpoint information.

```
USE master;
GO

IF NOT EXISTS (SELECT 1 FROM sys.endpoints WHERE name = 'Hadr_endpoint')
EXEC ('CREATE ENDPOINT Hadr_endpoint
STATE = STARTED
AS TCP (LISTENER_PORT = 5022)
FOR DATA_MIRRORING (
ROLE = ALL,
AUTHENTICATION = WINDOWS NEGOTIATE,
ENCRYPTION = REQUIRED ALGORITHM AES
)');
GO

IF NOT EXISTS (SELECT 1 FROM sys.server_principals WHERE name = 'databankenDEZ\sql')
CREATE LOGIN [databankenDEZ\sql] FROM WINDOWS;
GO

GRANT CONNECT ON ENDPOINT::Hadr_endpoint TO [databankenDEZ\sql];
GO

SELECT name, state_desc, port
FROM sys.tcp_endpoints
WHERE type_desc = 'DATABASE_MIRRORING';
```

100 % No issues found Ln: 25 Ch: 40 SPC CRLF

	name	state_desc	port
1	Hadr_endpoint	STARTED	5022

Deze query moet je ook uitvoeren op de Core server

Airport database Backup op de GUI Server en Restore op de CORE server

Open SSMS op Member01

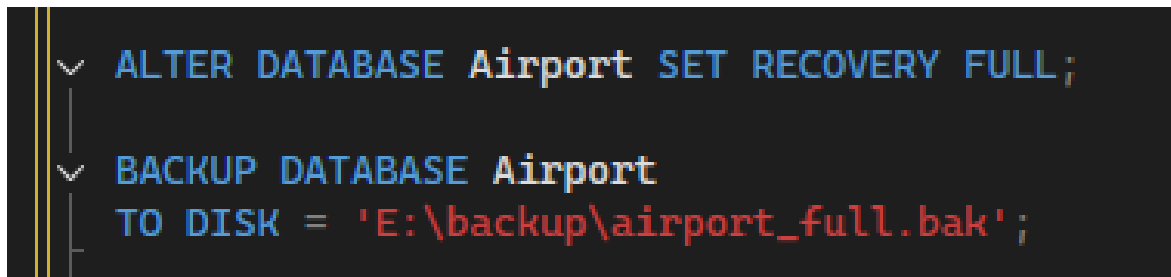
Open een nieuwe query.

- Zet database op Full recovery:

```
ALTER DATABASE Airport SET RECOVERY FULL;
```

- Neem een volledige backup (zorg ervoor dat de map bestaat):

```
BACKUP DATABASE Airport TO DISK='C:\backup\airport_full.bak';
```



```
✓ ALTER DATABASE Airport SET RECOVERY FULL;  
✓ BACKUP DATABASE Airport  
  TO DISK = 'E:\backup\airport_full.bak';
```

Maak ook een LOG backup:

```
BACKUP LOG Airport
```

```
TO DISK = 'E:\SQLBackups\Airport_log.bak'
```

```
WITH INIT;
```

Kopieer beide backup bestanden naar Member02:

Open Powershell in Member02 en voer onderstaande commandos uit:

```
Copy-Item "\\Member01DEZ\SQLBackups\Airport_full.bak" "K:\SQLBackups\"
```

```
Copy-Item "\\Member01DEZ\SQLBackups\Airport_log.bak" "K:\SQLBackups\"
```

Restore de backup op member02 in SQL:

```
RESTORE DATABASE Airport
```

```
FROM DISK = 'K:\SQLBackups\Airport_full.bak'
```

```
WITH
```

```
    MOVE 'Airport' TO 'K:\SQLData\Airport.mdf',
```

```
    MOVE 'Airport_log' TO 'K:\SQLLogs\Airport_log.ldf',
```

```
    NORECOVERY;
```

```
RESTORE LOG Airport
```

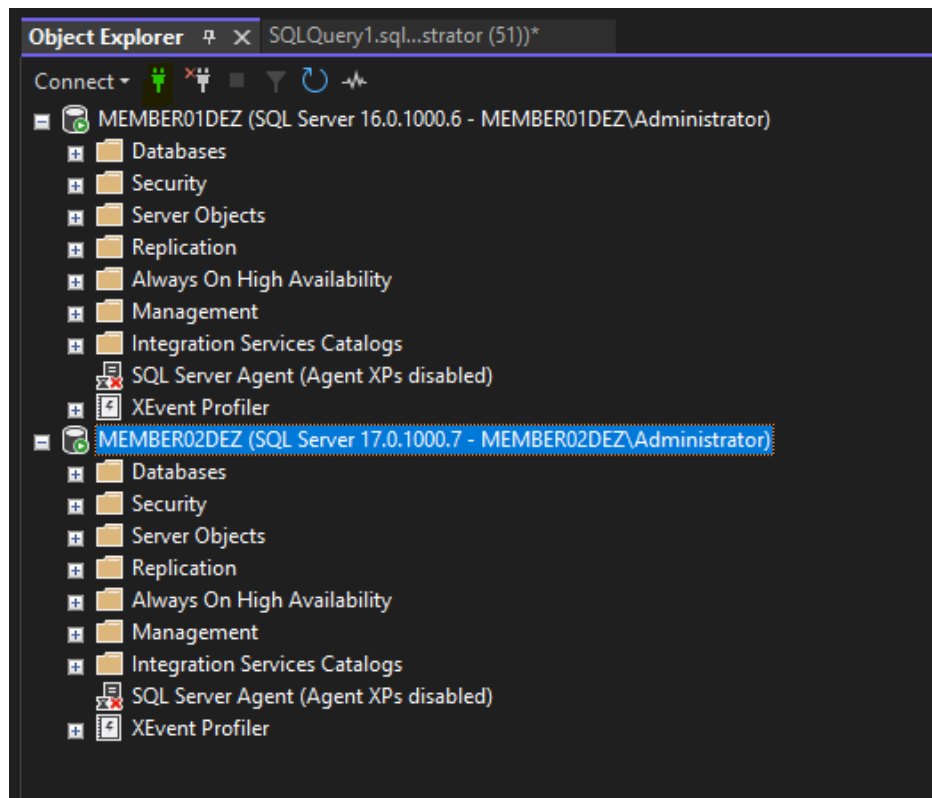
```
FROM DISK = 'K:\SQLBackups\Airport_log.bak'
```

```
WITH NORECOVERY;
```

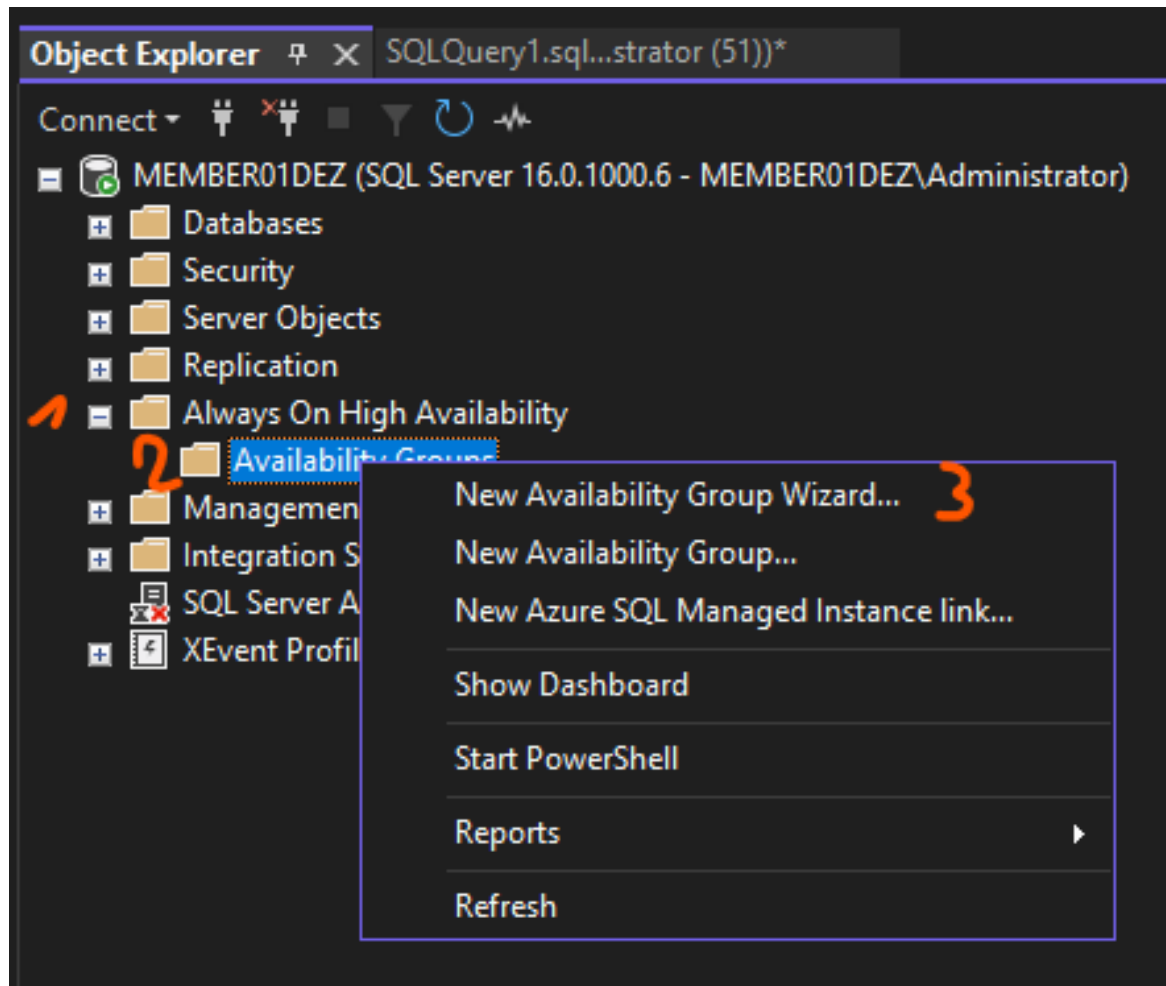
Aanmaken van de Availability Group op Member01

Zorg ervoor dat je connecteerd met de Core server door in de Object Explorer op de knop naast Connect te klikken en te connecteren met Windows Authentication.

Je zult dan zien dat de Core server wordt toegevoegd



Ga naar Always On High Availability → New Availability Group Wizard.



Nu zit je in de Wizard.

Naam: AirportAG

Cluster type: Windows Server Failover Cluster

New Availability Group



Specify Availability Group Options

[Introduction](#)
[Specify Options](#)
[Select Databases](#)
[Specify Replicas](#)
[Select Data Synchronization](#)
[Validation](#)
[Summary](#)
[Results](#)

[Help](#)

Specify availability group options

Availability group name:

Cluster type:

☐ Database Level Health Detection

☐ Per Database DTC Support

☐ Contained

☐ Reuse System Databases

Klik op Next.

Selecteer de database

Kies Airport database.

New Availability Group



Select Databases

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Select user databases for the availability group.

User databases on this instance of SQL Server:

	Name	Size	Status	Password
☐	AdventureWorks2019	336,0 MB	Full recovery mode is required	
☒	Airport	202,1 MB	Meets prerequisites	
☐	Employees	208,0 MB	Full backup is required	
☐	Oefeningen	16,0 MB	Full backup is required	
☐	Project	20,0 MB	Full backup is required	

Klik op Next.

Replicas toevoegen

Replica 1 (Primary – Member01DEZ):

- Synchronous Commit

Klik op “Add Replica” om de Core server toe te voegen:

Connect to Server

SQL Server

Login | Connection Properties | Always Encrypted | Additional Connection Parameters

Server

Server type: Database Engine

Server name: MEMBER02DEZ

Authentication: Windows Authentication

User name: MEMBER01DEZ\Administrator

Password:

☐ Remember password

Connection Security

Encryption: Mandatory

☒ Trust server certificate

Host name in certificate:

Connect Cancel Help Options <<

Replica 2 (Secondary – Member02DEZ):

-Automatic Failover aanvinken

- Synchronous Commit

Het moet er zo uit zien:

Specify an instance of SQL Server to host a secondary replica.

Replicas Endpoints Backup Preferences Listener Read-Only Routing

Availability Replicas:

Server Instance	Initial Role	Automatic Failover (Up to 5)	Availability Mode	Readable Secondary
Member01DEZ	Primary	☒ ✓	Synchronous commit ▾	No ▾
Member02DEZ	Secondary	☒ ✓	Synchronous commit ▾	No ▾

Endpoints configureren

Ga naar het volgende tabblad Endpoints

Hier moet je zelf niks aanpassen als je alles juist hebt gedaan.

- Endpoint poort moet 5022 zijn.

Specify an instance of SQL Server to host a secondary replica.

Replicas Endpoints Backup Preferences Listener Read-Only Routing

Endpoint values:

Server Name	Endpoint URL	Port Number	Endpoint Name	Encrypt Data	SQL Server Service Account
Member01DEZ	TCP://Member01DEZ.databankenDEZ.local:5022	5022	Hadr_endpoint	☒	DATABANKENDEZ\
Member02DEZ	TCP://Member02DEZ.databankenDEZ.local:5022	5022	Hadr_endpoint	☒	NT SERVICE\MSSQLSE

Listener configureren

- Naam: AirportListener

- Poort: 1433

- Statisch IP-adres toevoegen → Let op: dit moet een ip adres zijn die in hetzelfde netwerk zit maar die nog niet in gebruik is.

Replicas Endpoints Backup Preferences **Listener** Read-Only Routing

Specify your preference for an availability group listener that will provide a client connection point:

☐ Do not create an availability group listener now
You can create the listener later using the Add Availability Group Listener dialog.

☒ **Create an availability group listener** 1
Specify your listener preferences for this availability group.

Listener DNS Name: AirportListener 2

Port: 1433 3

Network Mode: Static IP

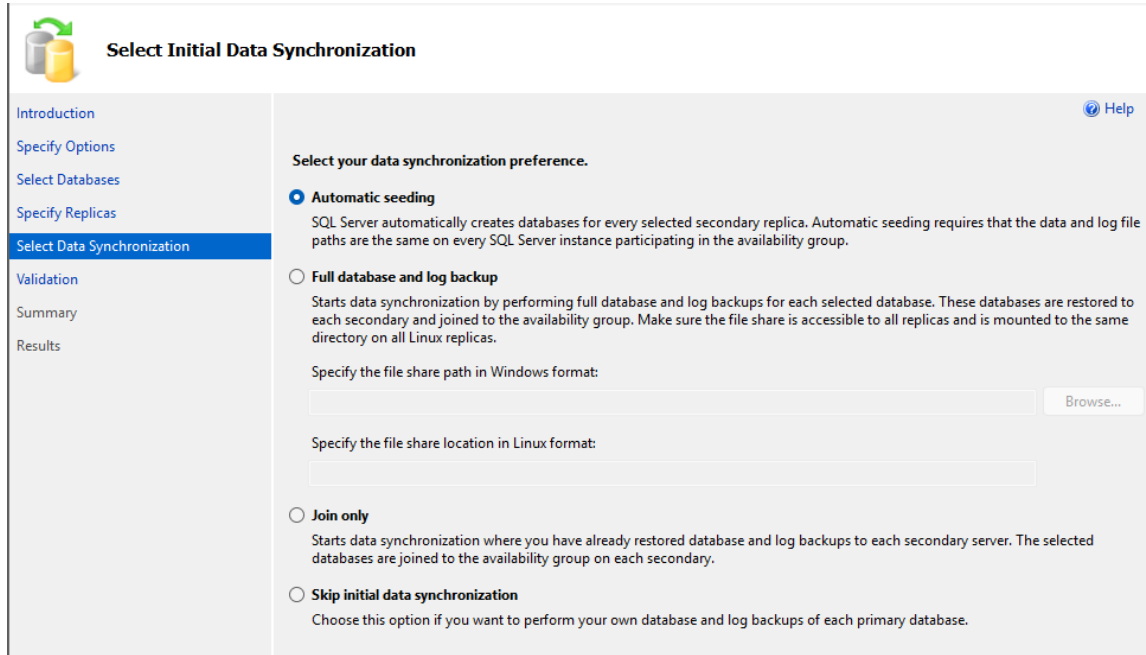
IP Address
172.25.87.100

4 Add... Remove

Klik dan op Next.

Data Synchronization

- Kies Automatic Seeding.



Select Initial Data Synchronization

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Select your data synchronization preference.

☒ **Automatic seeding**
SQL Server automatically creates databases for every selected secondary replica. Automatic seeding requires that the data and log file paths are the same on every SQL Server instance participating in the availability group.

☐ **Full database and log backup**
Starts data synchronization by performing full database and log backups for each selected database. These databases are restored to each secondary and joined to the availability group. Make sure the file share is accessible to all replicas and is mounted to the same directory on all Linux replicas.

Specify the file share path in Windows format:

Specify the file share location in Linux format:

☐ **Join only**
Starts data synchronization where you have already restored database and log backups to each secondary server. The selected databases are joined to the availability group on each secondary.

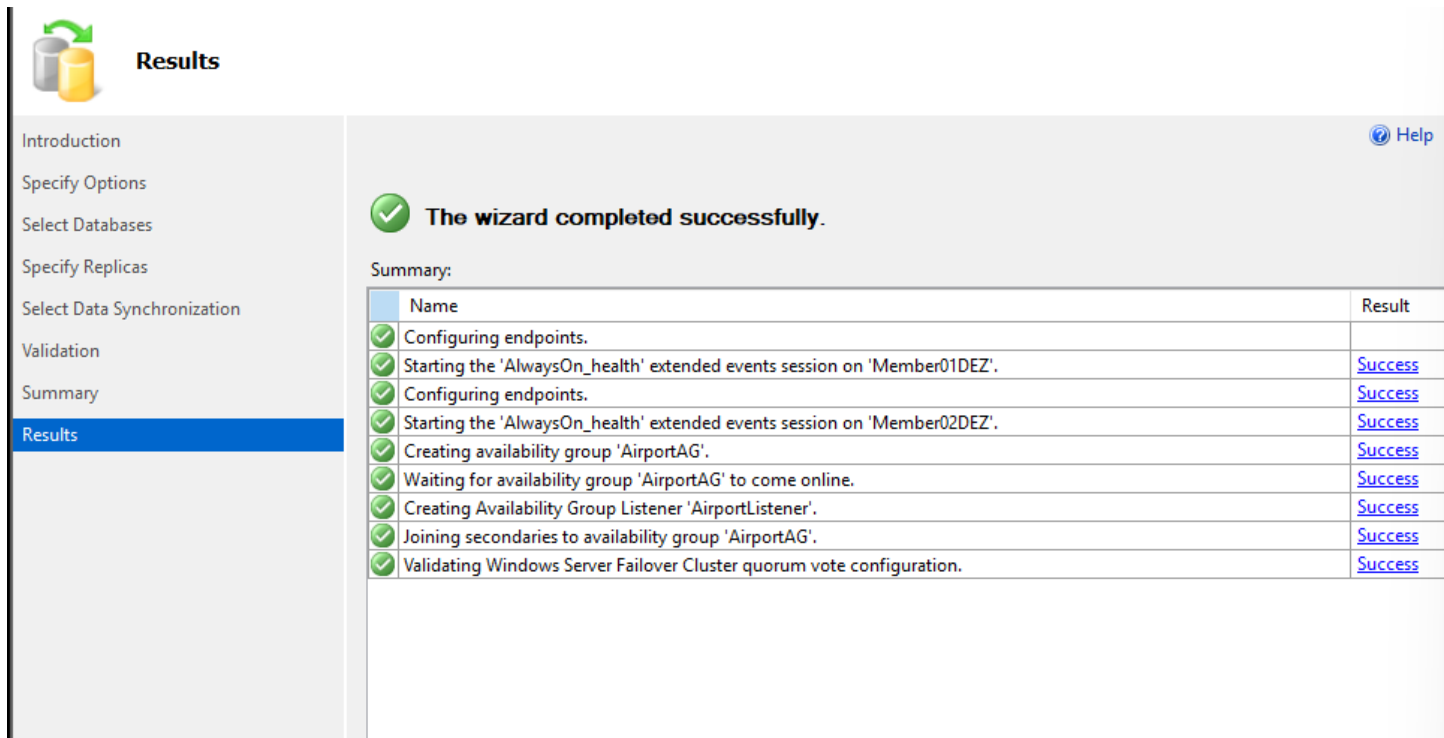
☐ **Skip initial data synchronization**
Choose this option if you want to perform your own database and log backups of each primary database.

Klik dan op Next

De Wizard zal nu afronden

Dit kan een paar minuten duren.

Ga verder als alles groen is en succesvol is aangemaakt.



Results

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 **The wizard completed successfully.**

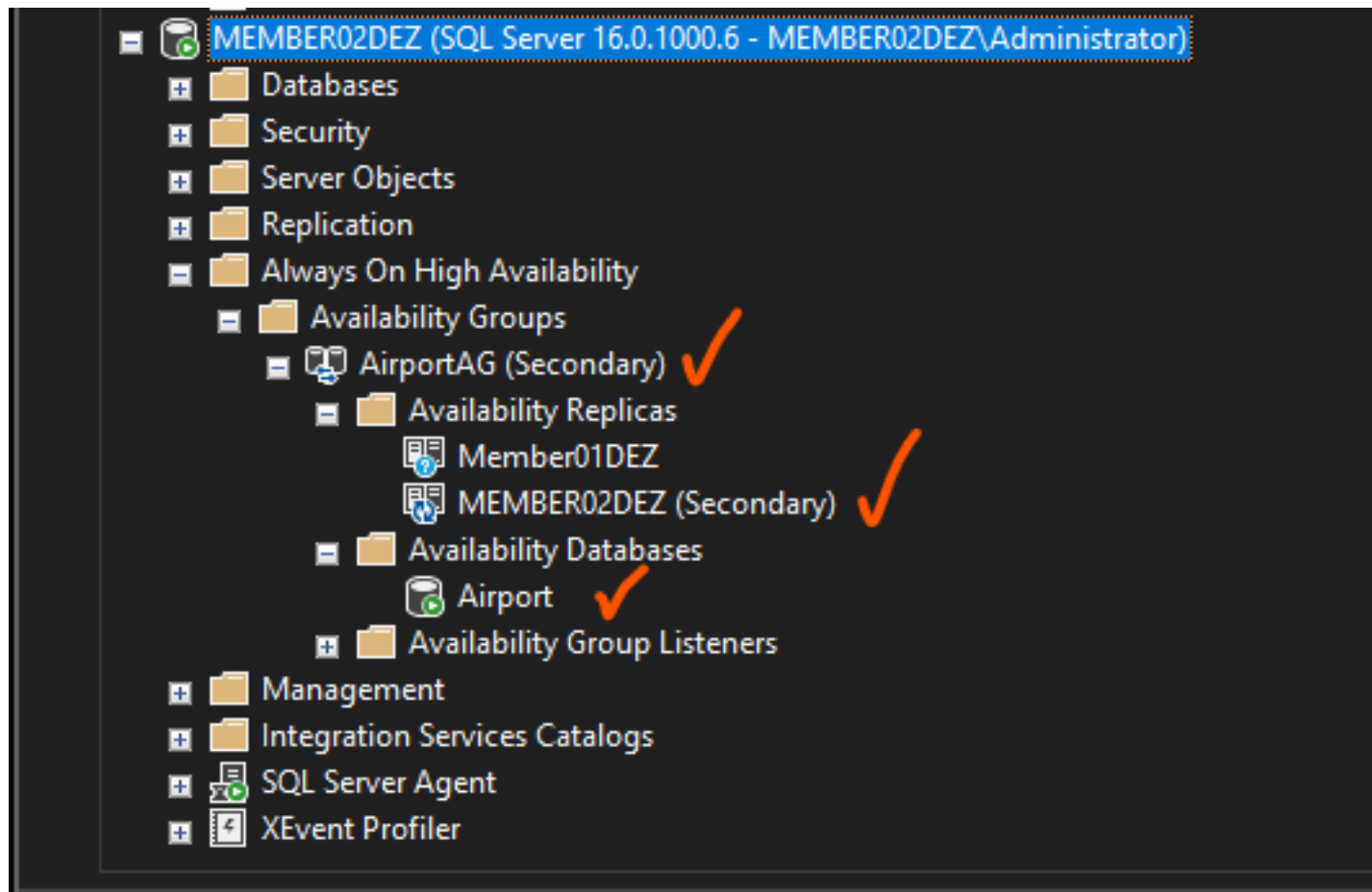
Summary:

Name	Result
 Configuring endpoints.	
 Starting the 'AlwaysOn_health' extended events session on 'Member01DEZ'.	Success
 Configuring endpoints.	Success
 Starting the 'AlwaysOn_health' extended events session on 'Member02DEZ'.	Success
 Creating availability group 'AirportAG'.	Success
 Waiting for availability group 'AirportAG' to come online.	Success
 Creating Availability Group Listener 'AirportListener'.	Success
 Joining secondaries to availability group 'AirportAG'.	Success
 Validating Windows Server Failover Cluster quorum vote configuration.	Success

Sluit dan af.

Test synchronisatie

Het moet eruit zien zoals onderstaande foto bij allebei de servers:



Query checks om uit te voeren voor controle synchronisatie op beide servers:

```
SELECT
    drs.database_id,
    drs.synchronization_state_desc,
    ars.role_desc,
    ars.connected_state_desc,
    ars.synchronization_health_desc
FROM sys.dm_hadr_database_replica_states drs
JOIN sys.dm_hadr_availability_replica_states ars
    ON drs.replica_id = ars.replica_id;
```



```

SELECT
    drs.database_id,
    drs.synchronization_state_desc,
    ars.role_desc,
    ars.connected_state_desc,
    ars.synchronization_health_desc
FROM sys.dm_hadr_database_replica_states drs
JOIN sys.dm_hadr_availability_replica_states ars
    ON drs.replica_id = ars.replica_id;

```

100 % 4 0 Ln: 17 Ch: 1

	database_id	synchronization_state_desc	role_desc	connected_state_desc	synchronization_health_desc
1	5	SYNCHRONIZED	SECONDARY	CONNECTED	HEALTHY
2	5	SYNCHRONIZED	PRIMARY	CONNECTED	HEALTHY

Correcte resulatoot = SYNCHRONIZED, CONNECTED & HEALTHY